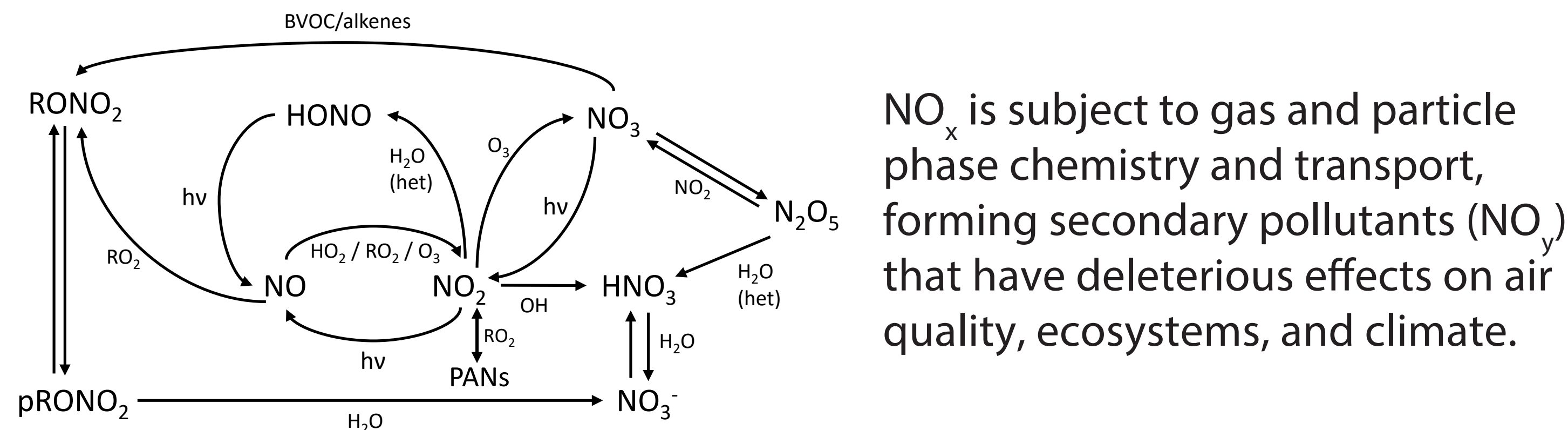


Motivation



Previous work¹⁻³ has characterized the distribution of NO_y in the atmosphere, however, there still exist many uncertainties in the relative importance of NO_y partitioning and loss pathways.

Methods

- We use v14.1.1 of the GEOS-Chem global chemical transport model (www.geos-chem.org), driven by Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) assimilated meteorology.
- Model simulations are conducted at a horizontal resolution of 2° x 2.5° and 47 hybrid sigma-pressure levels. The model includes a simulation of HO_x-NO_x-VOC-O₃-halogen chemistry^{4,5} coupled to aerosol thermodynamics.^{6,7}

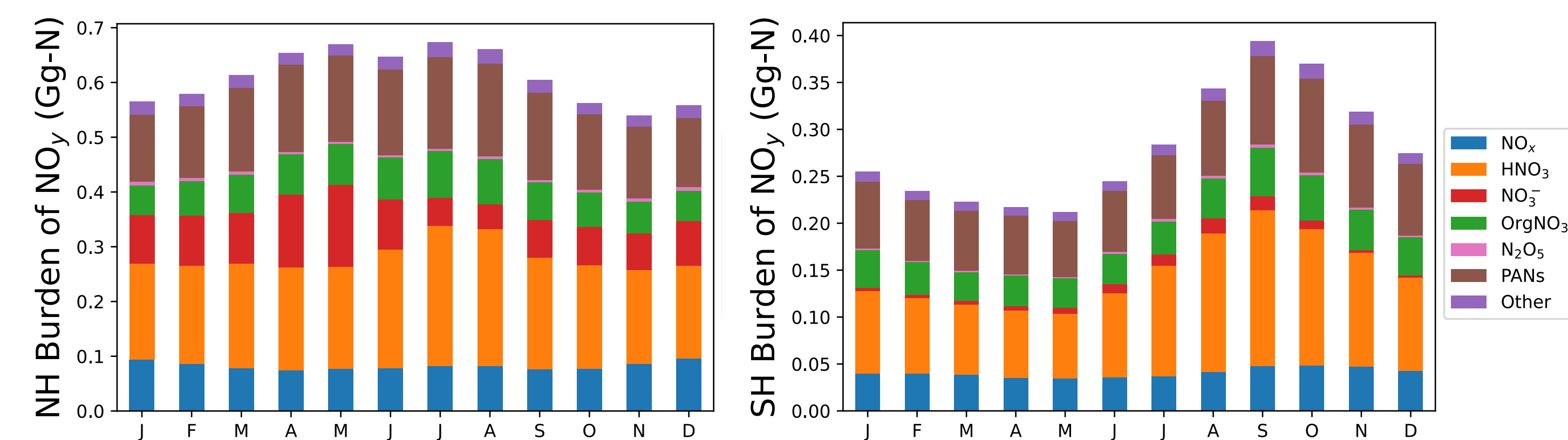


Figure 1: Modeled seasonal changes in the global burden of NO_y species in 2018

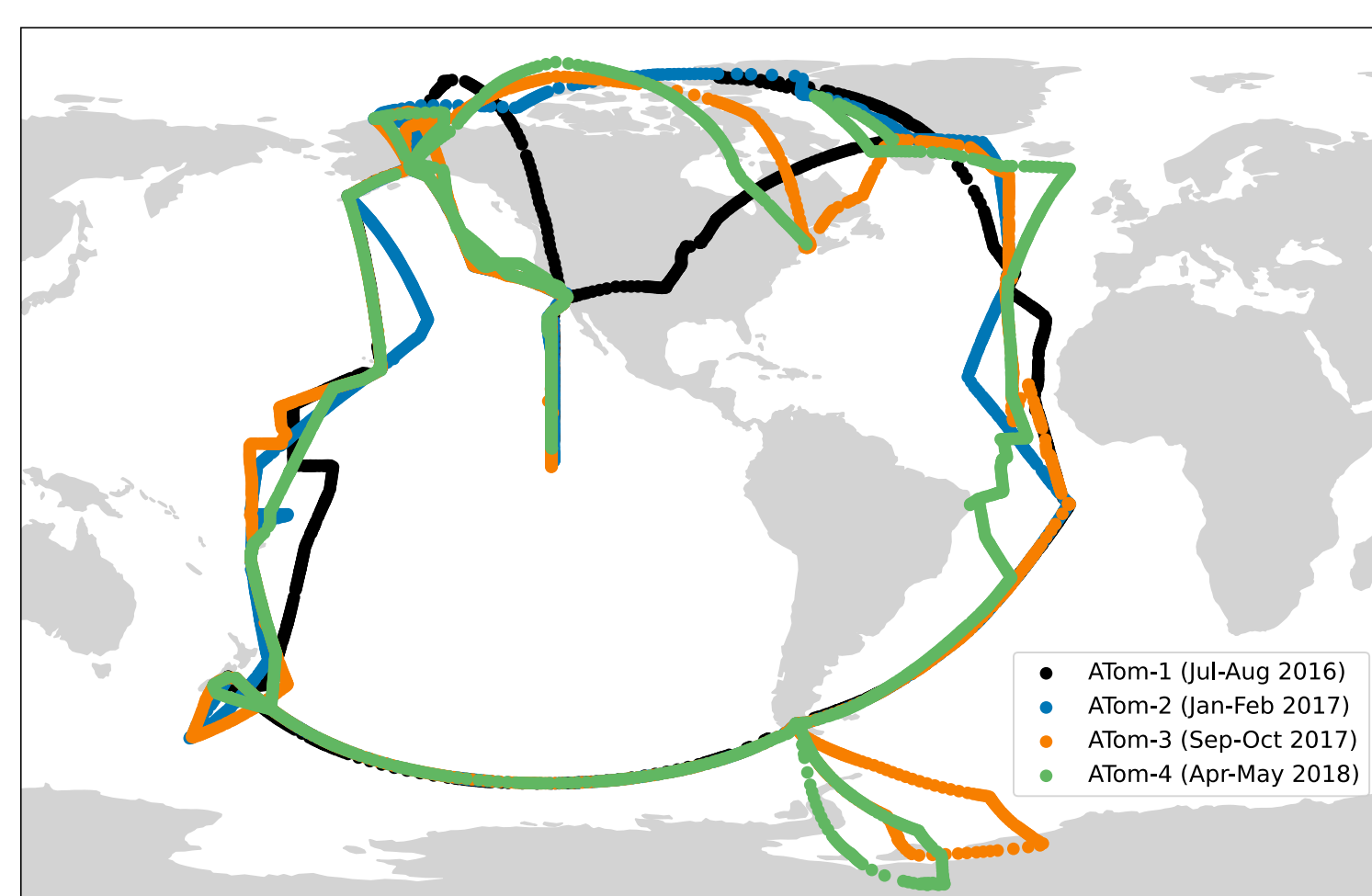


Figure 2: Coverage of ATom flights conducted between 2016 and 2018.

- We use airborne measurements (DC8) of NO_y made in the remote atmosphere during the Atmospheric Tomography Mission (ATom) to evaluate GEOS-Chem's representation of NO_y chemistry.

References

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- [5] X. Wang et al., *ACP* (2021)
- [6] Park et al., *JGR* (2004)
- [7] Pye et al., *JGR* (2009)
- [8] Luo et al., *GMD* (2019)
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- [10] Travis et al., *ACP* (2020)
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- [12] Fairlie et al., *Atm. Env.* (2010)
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- [15] Kasibhatla et al., *ACP* (2018)
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Model

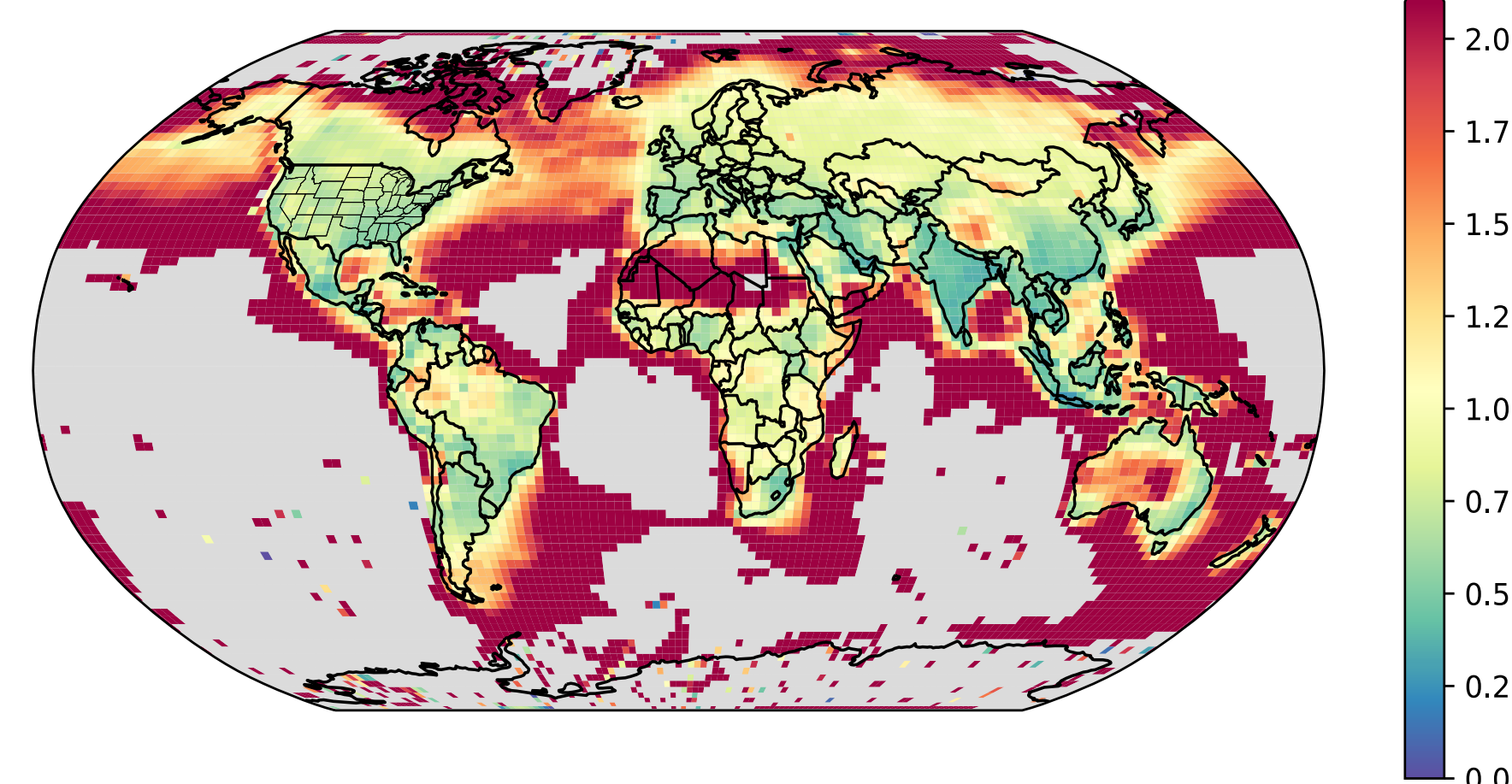


Figure 3: Modeled NO_x column chemical lifetime (in days).

- The mean global lifetime of NO_x is 21 hours, and lifetimes >5 days are masked out. NO_x lifetime is largely governed by the rate of oxidation of NO₂ to HNO₃ by OH and to PANs by RO₂ near the surface where NO_x burdens are highest.

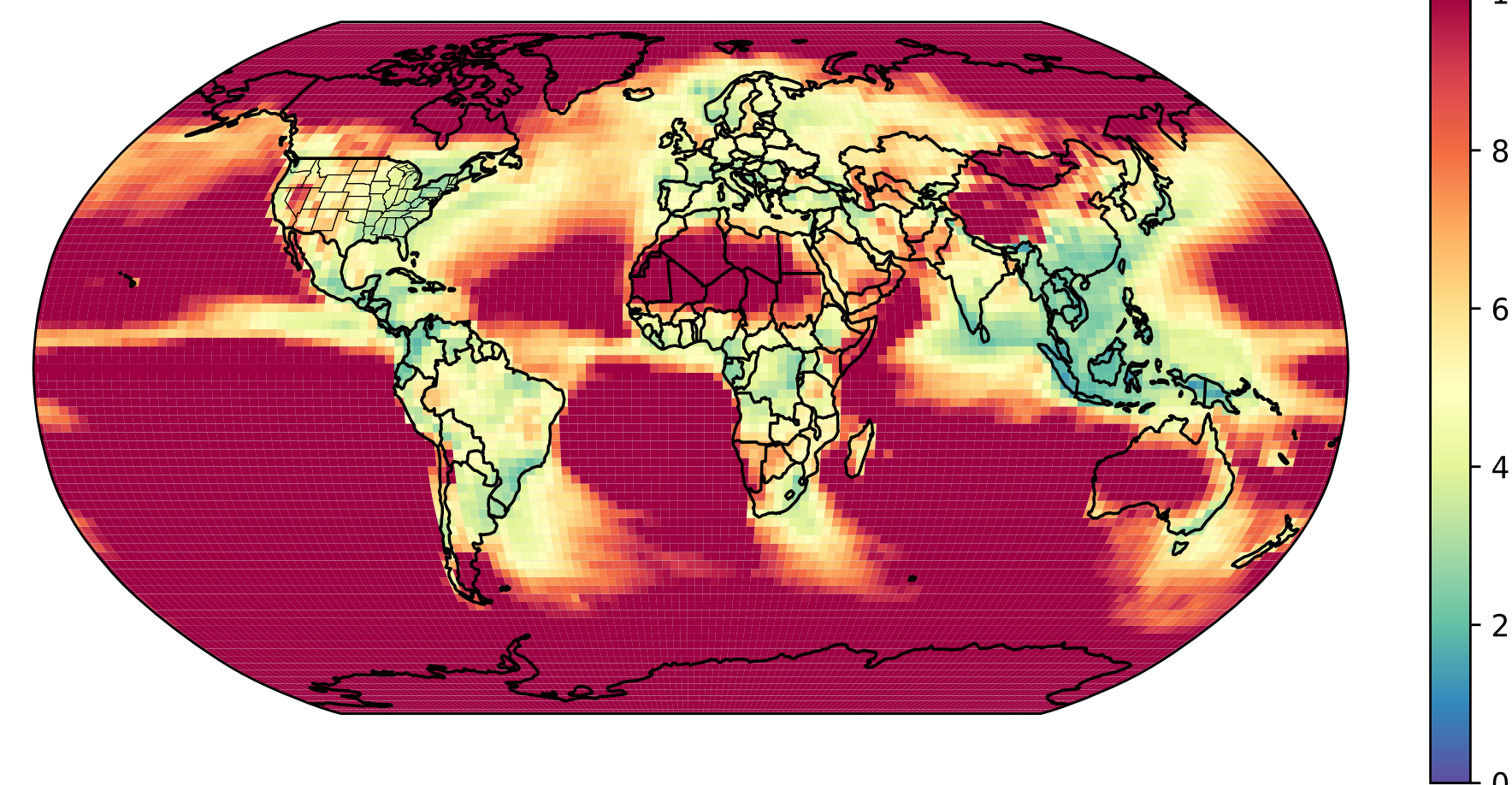


Figure 4: Modeled NO_y lifetime to deposition (in days). The mean global lifetime is 5.2 days, with approximately equal contributions from wet and dry deposition. Over half NO_y deposition is in the form of HNO₃.

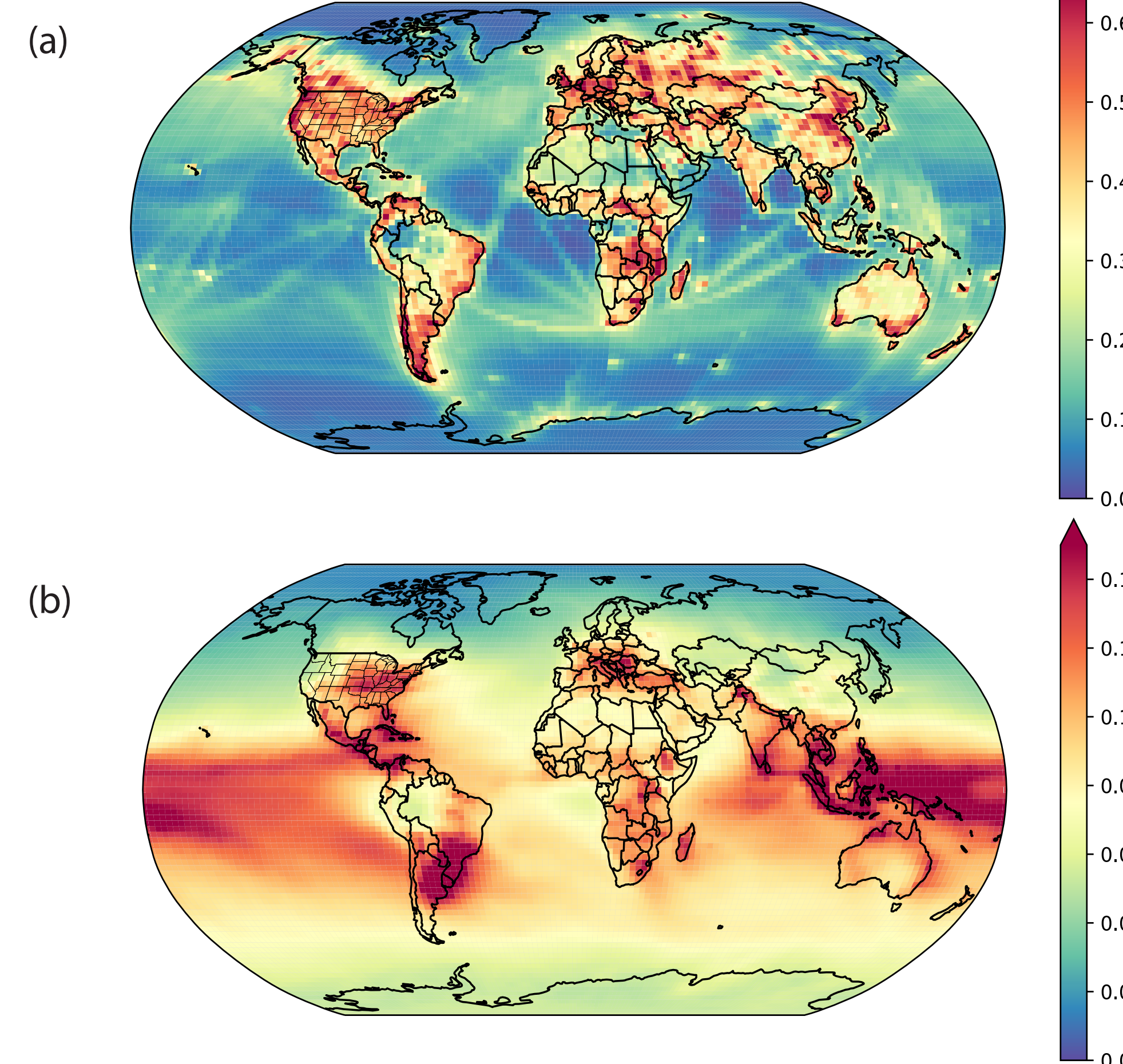


Figure 5: Modeled NO_x:NO_y at (a) the surface and (b) 500 hPa.

- High NO_x:NO_y regions are indicative of fresh NO_x emissions near the surface, with peaks near urban centers, major shipping lanes, and biomass burning regions.
- In the free troposphere, much of the available NO_x is consumed to form longer-lived NO_z (= NO_y - NO_x), and lower NO_x:NO_y ratios reflect mixing and transport.

Observations

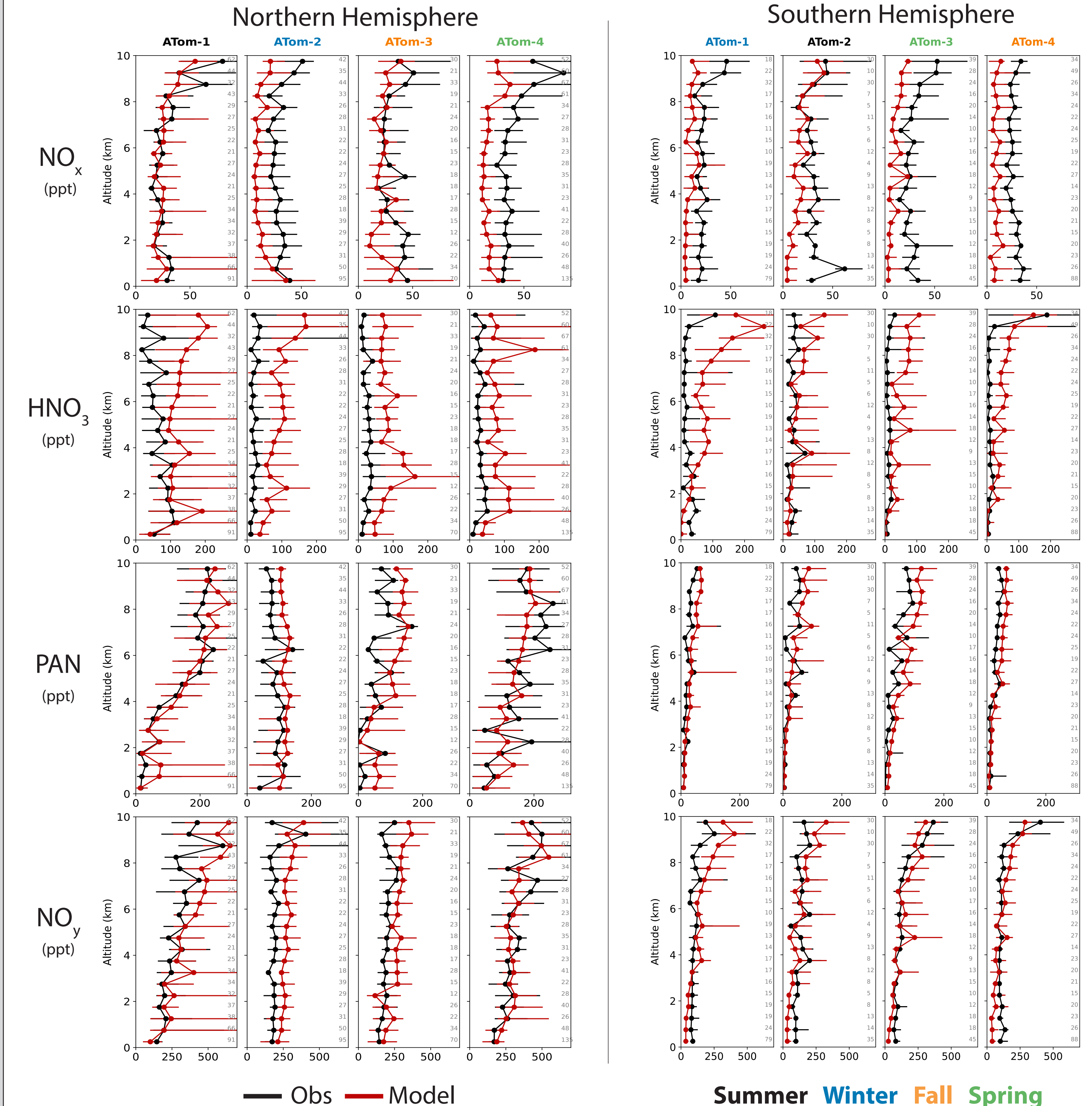


Figure 6: Modeled and observed (median) vertical profiles of NO_y species during ATom flights. Campaign label colors indicate hemispheric seasons across the mission.

- GEOS-Chem underestimates NO_x throughout the troposphere, across all seasons except boreal summer.
- The model systematically overestimates HNO₃, a long-standing uncertainty.
- An updated wet deposition scheme^{8,9} is able to address this high bias over the remote ocean.¹⁰ However these updates are inconsistent with continental measurements of deposition made over the United States.¹¹
- The model slightly overestimates NO_y, largely due to the overestimate of HNO₃, indicating either excess production or inadequate removal of NO_y.
- This high model bias in NO_y, taken with the low model bias in NO_x, leads to an overall low bias of NO_x:NO_y, with only boreal summer being well-represented.

Next Steps

- Evaluate model against the full suite of NO_y observations made by ATom.
- Track chemical production and loss of key NO_y species in GEOS-Chem.
- Evaluate sensitivity of NO_y budget to acidic uptake on dust,¹² aerosol nitrate photolysis.¹³⁻¹⁷